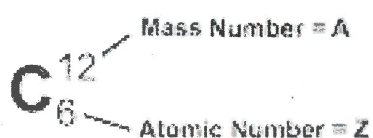
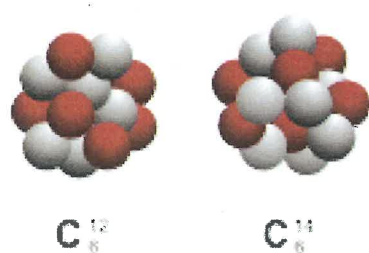
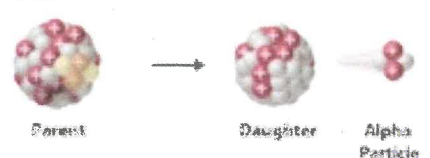
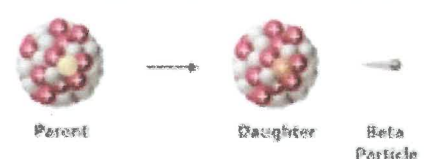
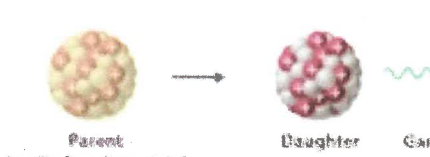


YEAR 9 PHYSICAL SCIENCE

ISOTOPES AND RADIOACTIVITY

PHYSICS BOOKLET

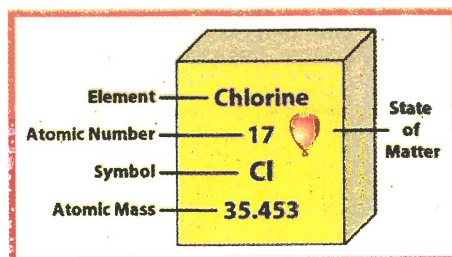


Decay Type	Radiation Emitted	Generic Equation	Model
Alpha decay	$^4_2\alpha$	$^A_ZX \longrightarrow ^{A-4}_{Z-2}X' + ^4_2\alpha$	 Parent Daughter Alpha Particle
Beta decay	$^0_{-1}\beta$	$^A_ZX \longrightarrow ^A_{Z+1}X' + ^0_{-1}\beta$	 Parent Daughter Beta Particle
Gamma emission	$^0_0\gamma$	$^A_ZX^* \xrightarrow{\text{Relaxation}} ^A_ZX' + ^0_0\gamma$	 Parent (excited nuclear state) Daughter Gamma ray

NAME: _____

TEACHER: _____

ISOTOPES & RADIOACTIVE DECAY



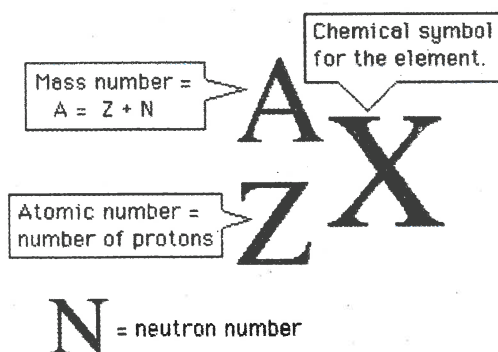
To review

- Atomic number = number of _____
- Number of protons = number of _____ (in a neutral atom)
- Number of _____ = atomic mass – atomic number

Atomic mass

- Atomic mass unit is a unit used to compare the _____ of atoms
- has the symbol _____
- 1 amu is approximately equal to the mass of a single _____ or _____.
- Based on _____.
- Atomic mass is the weighted _____ of all the naturally occurring _____ of that _____.

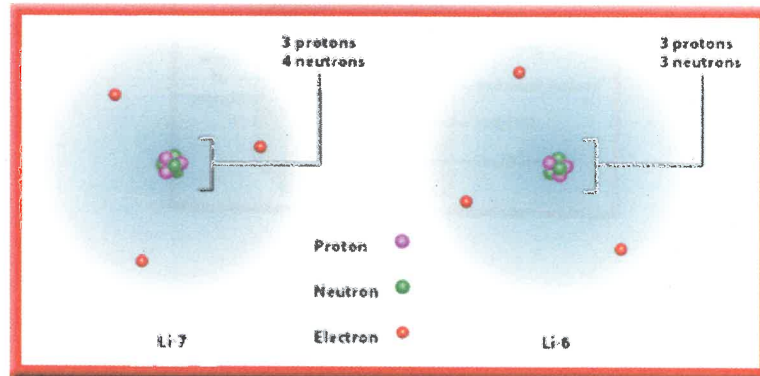
Notation:



Other ways to write elements:

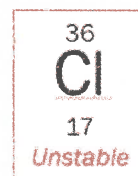
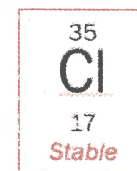
Isotopes:

- Isotopes of an element have _____ because they have different numbers of _____, but they have the same _____.
- Isotopes of lithium:



Why are some isotopes unstable?

- Wrong _____ of protons and neutrons:
 - Too many _____ for the number of _____
 - Too many _____ for the number of _____
- _____ protons and neutrons (the _____ is _____)
- Stable or unstable?
 - Stable isotopes:
 - Stable _____
 - Right balance of _____ and _____
 - Not too big
 - Unstable isotopes:
 - _____ nuclei
 - _____



What do isotopes do when they are unstable?

- They _____ (_____)
- When they break down they _____ or _____ from their nucleus (called _____)
- This turns them into a _____

Isotopes

Aims

- 1 To know that isotopes are atoms of the same element that have the same number of protons but a different number of neutrons.
- 2 To work out the atomic structure of various elements.

Questions

The following questions are about neutrons and isotopes.

- 1 Where in an atom are the neutrons?
- 2 What is the relative mass of a neutron?
- 3 Complete the equation to show how to work out the number of neutrons in an atom from the information in the periodic table
..... - = number of neutrons
- 4 Underline the statement that correctly describes an isotope.
 - a) An atom with different particles in the nucleus.
 - b) An atom of the same element with different numbers of neutrons.
 - c) An atom of the same element with neutrons and protons different.
- 5 How many neutrons are there in these isotopes?
 - a) $^{35}_{17}\text{Cl}$
 - b) $^{37}_{17}\text{Cl}$
- 6 Isotopes of the same element have the same physical properties. True or false?
- 7 Match up the beginnings with the endings to make three complete sentences.
 - a) Almost all of the mass of an atom is found the mass number.
 - b) We would say an unstable isotope is in the nucleus.
 - c) The number of neutrons and protons gives us radioactive.
- 8 Complete the table with information about the isotopes of uranium.

Isotope	Number of protons	Number of electrons	Number of neutrons
Uranium-235 $^{235}_{92}\text{U}$			
Uranium-237 $^{237}_{92}\text{U}$			

Atomic structure: Worksheet 4.1.3

1 Key terms

Give a definition of each of the following terms.

1. Mass number
2. Atomic number
3. Nucleon number
4. Isotope
5. Ion

2 Using symbols to represent atoms

Complete the table.

Name of element	Chemical symbol	Mass number (A)	Atomic number (Z)	Number of protons	Number of neutrons	Number of electrons
Carbon		12		6	6	
	Li					
Magnesium						12
	U	235	92			
Radium						
			26			26

3 Ionisation

Complete the table.

Formula of ion	Number of protons	Number of neutrons	Number of electrons
Na ⁺			
Ca ²⁺			
F ⁻			
O ²⁻			

Calculating the relative atomic mass

Isotopes have the same number of _____ but different numbers of _____.

The relative atomic mass is the average mass of an element. To calculate the relative atomic mass, you need to know the mass of the isotope and how much there is of each isotope (abundance).

$$\text{RAM} = \frac{(\text{mass number} \times \text{percentage}) \text{ of isotope 1} + (\text{mass number} \times \text{percentage}) \text{ of isotope 2}}{100}$$

1. The table shows information about the isotopes of an element.

	Mass number	Percentage (%) abundance
Isotope 1	10	20
Isotope 2	11	80

Calculate the relative atomic mass (A_r) of the element. Give your answer to 1 decimal place.

$$\text{RAM} = \frac{(10 \times \underline{\quad}) + (11 \times \underline{\quad})}{100} = \frac{\underline{\quad} + \underline{\quad}}{100} =$$

2. Calculate the relative atomic mass of bromine. Give your answer to 1 decimal place.

	Mass number	Percentage (%) abundance
^{79}Br	79	50.7
^{81}Br	81	49.3

$$\text{RAM Bromine} = \frac{(79 \times \underline{\quad}) + (81 \times \underline{\quad})}{100} = \frac{\underline{\quad} + \underline{\quad}}{100} =$$

3. Calculate the relative atomic mass of gallium. Give your answer to 1 decimal place.

	Mass number	Percentage (%) abundance
^{69}Ga	69	60.1
^{71}Ga	71	39.9

$$\text{RAM Gallium} = \frac{(69 \times \underline{\quad}) + (71 \times \underline{\quad})}{100} = \frac{\underline{\quad} + \underline{\quad}}{100} =$$

4. Calculate the relative atomic mass of nitrogen. Give your answer to 3 decimal places.

	Mass number	Percentage (%) abundance
^{14}N		99.6
^{15}N		0.4

$$\text{RAM Nitrogen} = \frac{(\underline{\quad} \times \underline{\quad}) + (\underline{\quad} \times \underline{\quad})}{100} =$$

5. Calculate the relative atomic mass of oxygen. Give your answer to 3 decimal places.

	Mass number	Percentage (%) abundance
^{16}O		99.8
^{18}O		0.2

$$\text{RAM Oxygen} = \frac{(\underline{\quad} \times \underline{\quad}) + (\underline{\quad} \times \underline{\quad})}{100} =$$



RELATIVE ATOMIC MASS

The relative atomic mass (A_r) of atoms is the average mass of all the different isotopes of an element (taking into account the amount of each isotope) on a scale where ^{12}C atoms have a mass of exactly 12.

Imagine you have 90 balls with mass 200 g, and 10 balls with mass 300 g. The average mass of the balls is given by:

$$\text{Average mass of balls} = \frac{\text{total mass of all the balls}}{\text{total number of balls}} = \frac{[(90 \times 200) + (10 \times 300)]}{90 + 10} = \frac{21000}{100} = 210 \text{ g}$$

The relative atomic mass of atoms is worked out in a similar way:

$$\text{Relative atomic mass } (A_r) = \frac{\text{total mass of all atoms}}{\text{total number of atoms}}$$

Element	Isotopes	Abundance	Relative atomic mass (A_r) (to 3sf)
Chlorine	$^{35}_{17}\text{Cl}$ $^{37}_{17}\text{Cl}$	75.8% 24.2%	$A_r = \frac{[(35 \times 75.8) + (37 \times 24.2)]}{75.8 + 24.2} = \frac{3548.4}{100} = 35.5 \text{ (3sf)}$
Lithium	^6_3Li ^7_3Li	7.6% 92.4%	
Bromine	$^{79}_{35}\text{Br}$ $^{81}_{35}\text{Br}$	50.7% 49.3%	
Copper	$^{63}_{29}\text{Cu}$ $^{65}_{29}\text{Cu}$	69.2% 30.8%	
Fluorine	$^{19}_9\text{F}$	100.0%	
Magnesium	$^{24}_{12}\text{Mg}$ $^{25}_{12}\text{Mg}$ $^{26}_{12}\text{Mg}$	79.0% 10.0% 11.0%	
Iron	$^{54}_{26}\text{Fe}$ $^{56}_{26}\text{Fe}$ $^{57}_{26}\text{Fe}$ $^{58}_{26}\text{Fe}$	5.8% 91.8% 2.1% 0.3%	
Krypton	$^{78}_{36}\text{Kr}$ $^{80}_{36}\text{Kr}$ $^{82}_{36}\text{Kr}$ $^{83}_{36}\text{Kr}$ $^{84}_{36}\text{Kr}$ $^{85}_{36}\text{Kr}$	0.4% 2.3% 11.6% 11.5% 57.0% 17.3%	

Radioactivity Questions

- 1 The figure below shows how Henri Becquerel discovered radioactivity in 1896. A packet of uranium salts was placed on top of a wrapped photographic plate. A key was between the container and the glass plate. When the plate was developed later, an image of the key was seen on it.



- a) Which of the following materials stopped the radiation from the uranium?

air glass metal paper

- b) Complete the following sentences using words from the list.

around through under

- i) The radiation from the uranium salts affected the plate _____ the key.
- ii) The radiation from the uranium salts did not affect the plate _____ the key.
- iii) The radiation from the uranium salts passed _____ the wrapper.

- 2 Complete the following sentences using words from the list.

atom(s) electron(s) neutron(s) nucleus (or nuclei) proton(s)

- a) The _____ of an _____ is composed of _____ and _____.
- b) The _____ in an _____ move about in the space outside the _____.
- c) A _____ is a positively charged particle with about the same mass as a _____ which is uncharged.
- d) An _____ has an equal and opposite charge to the _____ and has a much smaller mass.

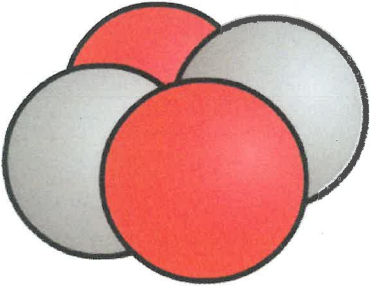
- 3 A scientist used a portable Geiger counter outdoors to test some rocks for radioactivity.

- a) The Geiger counter 'clicked' more when it was held near a certain rock than when it was held near the other rocks. What conclusion can you draw about the rock that made it click more?
- b) She noticed that the clicks were unpredictable. Why was this?

Name:

Class:

Date:

	radiation	<u>Other properties/notes:</u>
<u>Number of protons:</u>	<u>Electrical charge:</u>	
<u>Number of neutrons:</u>	<u>Can be 'stopped' by:</u>	
<u>Symbol (in form $^A_Z X$):</u>		

	radiation	<u>Other properties/notes:</u>
<u>Number of protons:</u>	<u>Electrical charge:</u>	
<u>Number of neutrons:</u>	<u>Can be 'stopped' by:</u>	
<u>Symbol (in form $^A_Z X$):</u>		

	radiation	<u>Other properties/notes:</u>
<u>Number of protons:</u>	<u>Electrical charge:</u>	
<u>Number of neutrons:</u>	<u>Can be 'stopped' by:</u>	
<u>Symbol (in form $^A_Z X$):</u>		

Alpha decay

- Two _____ and two _____ are lost from a nucleus when it emits an _____. This means that:
 - The atomic mass number _____ by _____
 - The atomic number _____ by _____
 - A _____ is formed that is _____ in the Periodic Table than the original element.

Alpha decay equation:

Beta decay

- In beta decay, a _____ changes into a _____ plus an _____. The proton stays in the nucleus. The _____ leaves the atom with high energy as a _____.
- The nucleus has _____ and _____. This means that:
 - The atomic _____ stays the same
 - The atomic number _____ by _____

Beta decay equation:

Gamma decay

- An unstable _____ kicks out a _____ and releases some of its _____
- The atom's _____ stays the _____, it just loses energy

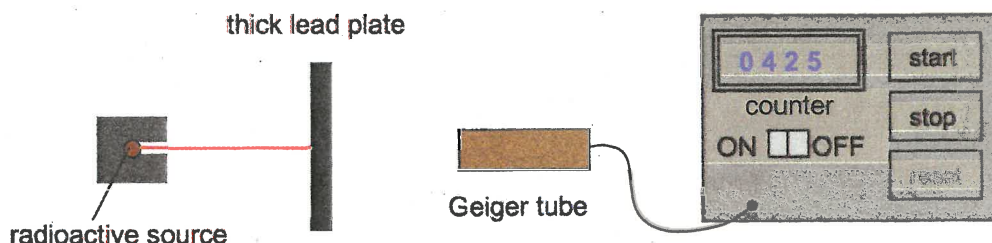
Gamma decay equation:

TO REVIEW (OPTIONS)

1. Read chapter 6.2 of Science Quest 9 (using class set)
2. Answer Understanding & Inquiring questions 1-11

A Radioactivity Puzzle

4. The figure below shows a Geiger tube used to detect radiation from a radioactive source.



The following measurements were made using the apparatus above.

	Counts in 300 seconds			
	Readings			Average
1 No source present	102	94	110	
2 Source present at fixed distance from tube				
a) No lead plate present	3466	3420	3410	
b) Thick lead plate present	105	109	89	
c) Aluminium sheet in place of the thick lead sheet	1834	1787	1818	

a) Complete the table by calculating the average readings.

b) Why are the readings on each line not the same?

c) What can you say from the table about the effect on the radiation of:

i) the lead plate? _____

ii) the aluminium plate? _____

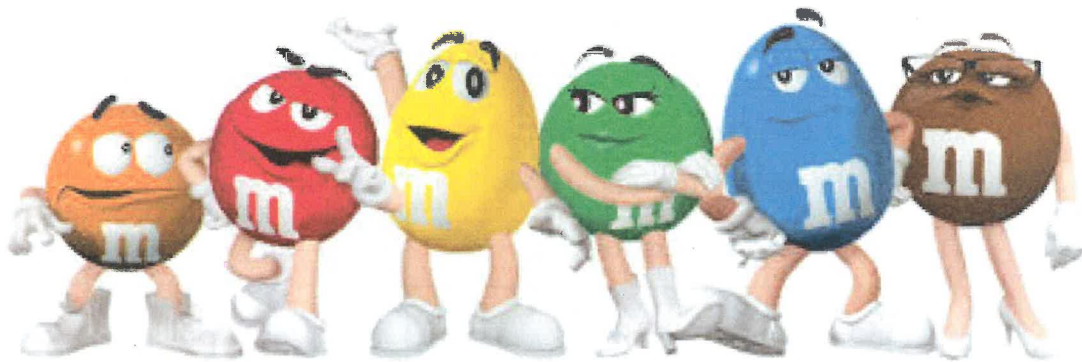
e) Why is it possible to say from the readings that:

i) gamma radiation is emitted by the source? _____

ii) alpha and beta radiation might be emitted by the source? _____

f) What further tests could you make using this arrangements to find out whether or not the source emits alpha radiation?

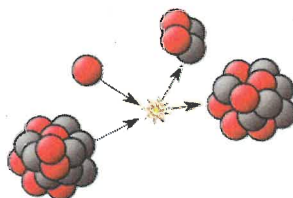
M&M Half Life Lab



Purpose: To simulate the decay of a radioactive isotope over time and to graph the data and relate it to radioactive decay and half-lives.

Pre-lab questions:

1. Define the term radioactive half-life



2. Why are M&Ms used for this lab rather than, say, Smarties?

3. What do the M and "blank" sides of the M&Ms represent?

Risk assessment:

M&Ms must only be handled if wearing gloves. Any that land on the table or floor must be discarded. Those that remain at the end of the task may be consumed once you have left the lab. If you have a latex allergy be careful to use the latex-free gloves, or to allow your partners to handle the M&Ms

Materials:

- Approx. 50 M&Ms, 1 paper towel/plate, 2 cups, gloves

Procedure:

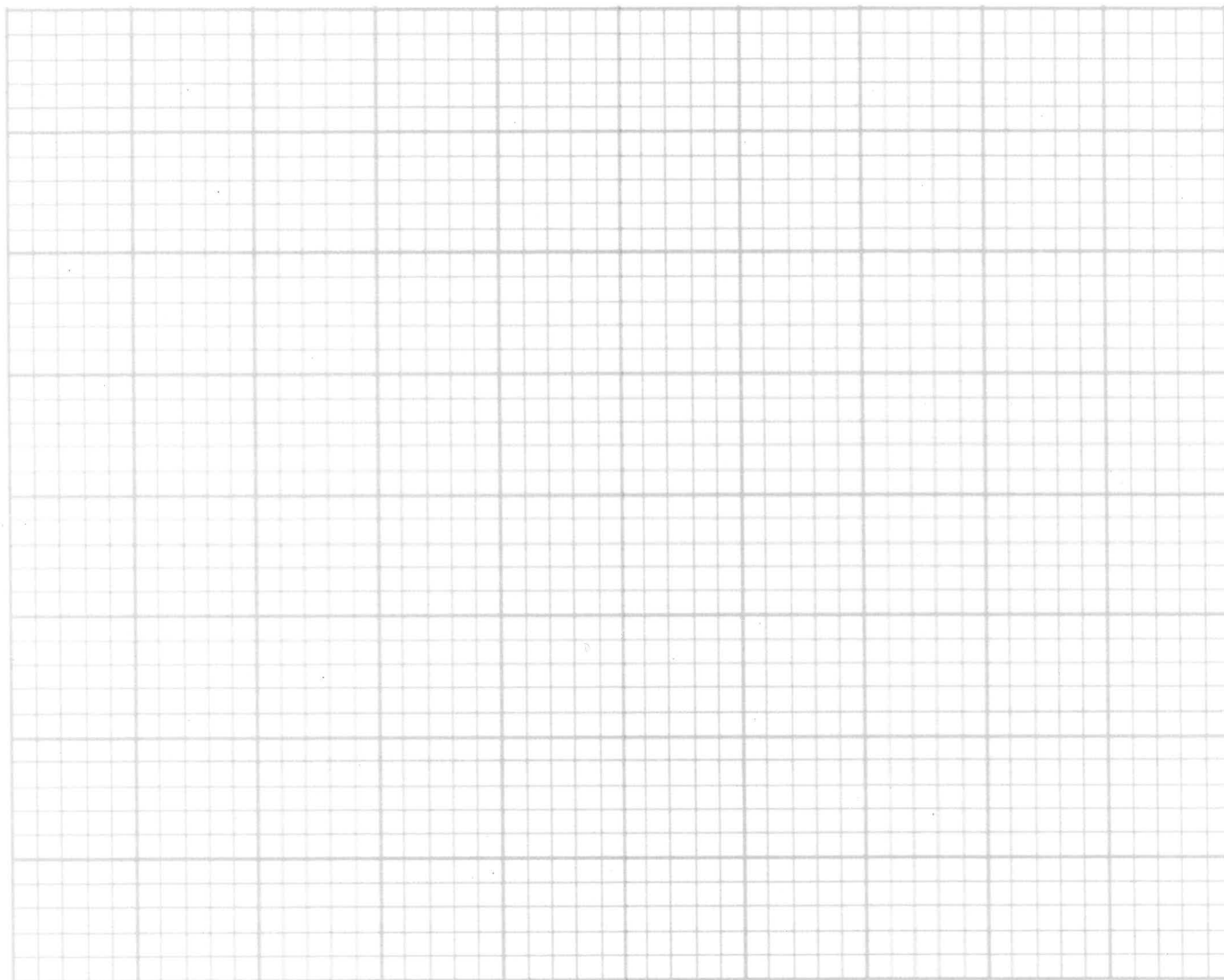
1. Place all M&Ms on the plate with the M facing up. These are the parent radioisotopes. Count how many M&Ms there are and write that in the table next to time zero.
2. Place M&Ms in cup and roll onto the paper towel.
3. Remove any M&Ms facing down. These will represent atoms of the isotope that **have** decayed. These can go in a second container for the remainder of the task.
4. Count the remaining face up M&Ms and record in your data table.
5. Repeat steps 3-5 until all of your M&Ms have 'decayed'. Add the count each time to the data table.
6. Repeat steps 1-6 for a second trial then a third trial.
7. Draw an X-Y scatter graph (by hand) on the graph paper provided.

Data Table:

Half-Life ($t_{1/2}$)	Number of "radioactive" M&Ms remaining			Experimental Remaining		Theoretical Remaining	
	Trial 1	Trial 2	Average	%	Fraction	%	Fraction
0				100	1	100	1
1						50	1/2
2						25	1/4
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							

Graph:

Once you have all your data points in your table, you need to draw a half-life vs average number of radioactive m&ms remaining graph below. Once you have all your data points on your graph, draw a free hand *line of best fit* (it should be a curve). *****Remember to include axes labels and a descriptive title*****



Describe your using the claim, evidence, reasoning (CER) model.

Using your knowledge and additional research explain how the half-lives of different isotopes are similar and how they are different.

Explain the ways the M&M model is useful in describing the decay of radioactive isotopes, and how it is limited.

Post-Lab Questions:

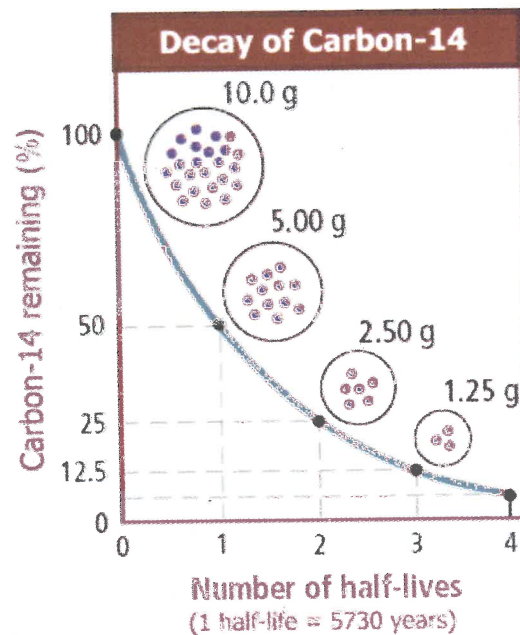
1. If 50% of a radioactive element remains after 4000 years, what is the half-life?
2. The half-life of a certain radioactive element is 1,250 years. What percentage of the atoms remain after 7,500 years?
3. The half-life of a certain radioactive element is 800 years. How old is an object if only 12.5% of the radioactive atoms in it remain?

Half life:

- The time taken for _____ of the original _____ of _____ to decay into their _____
- Is different for all radioactive substances
 - uranium-238 half life = _____
 - _____ half life = 22 minutes

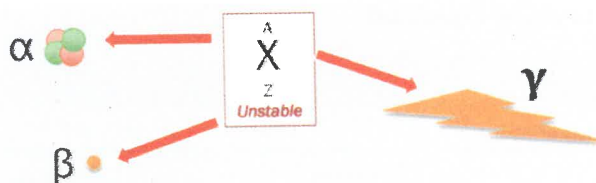
Half-life graph of carbon-14

- _____ decays into _____
- In _____ years, half the carbon-14 _____ in a sample will have _____ into nitrogen-14

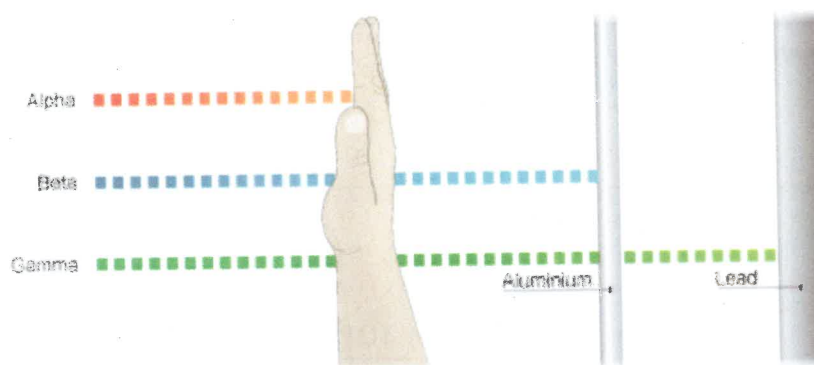


Radiation

- Three types:
 - _____ particles _____
 - _____ particles _____
 - _____ rays _____



Penetrating power of radiation

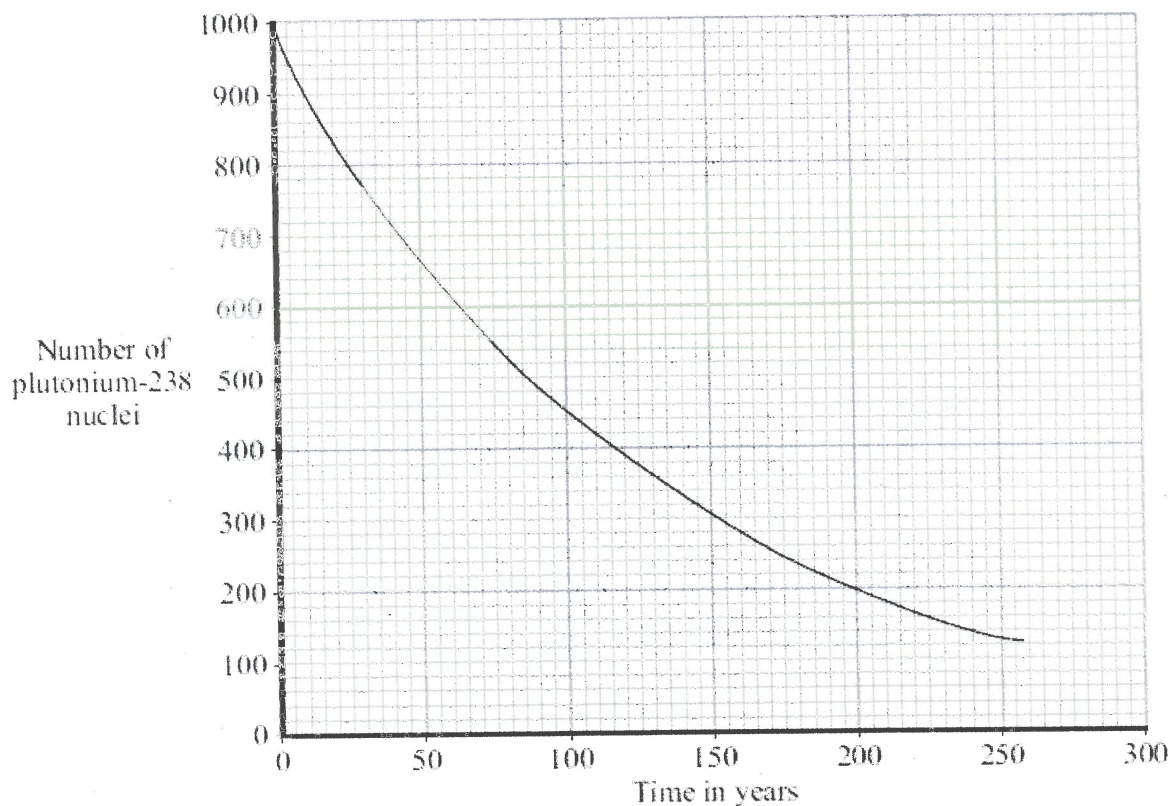


Ionising power of radiation

- The level of _____ done by the type of radiation
- Causes _____ to the _____ in a substance
- _____ has the _____ ionising power therefore it can do the _____ damage
- _____ is _____ and _____ radiation is the _____

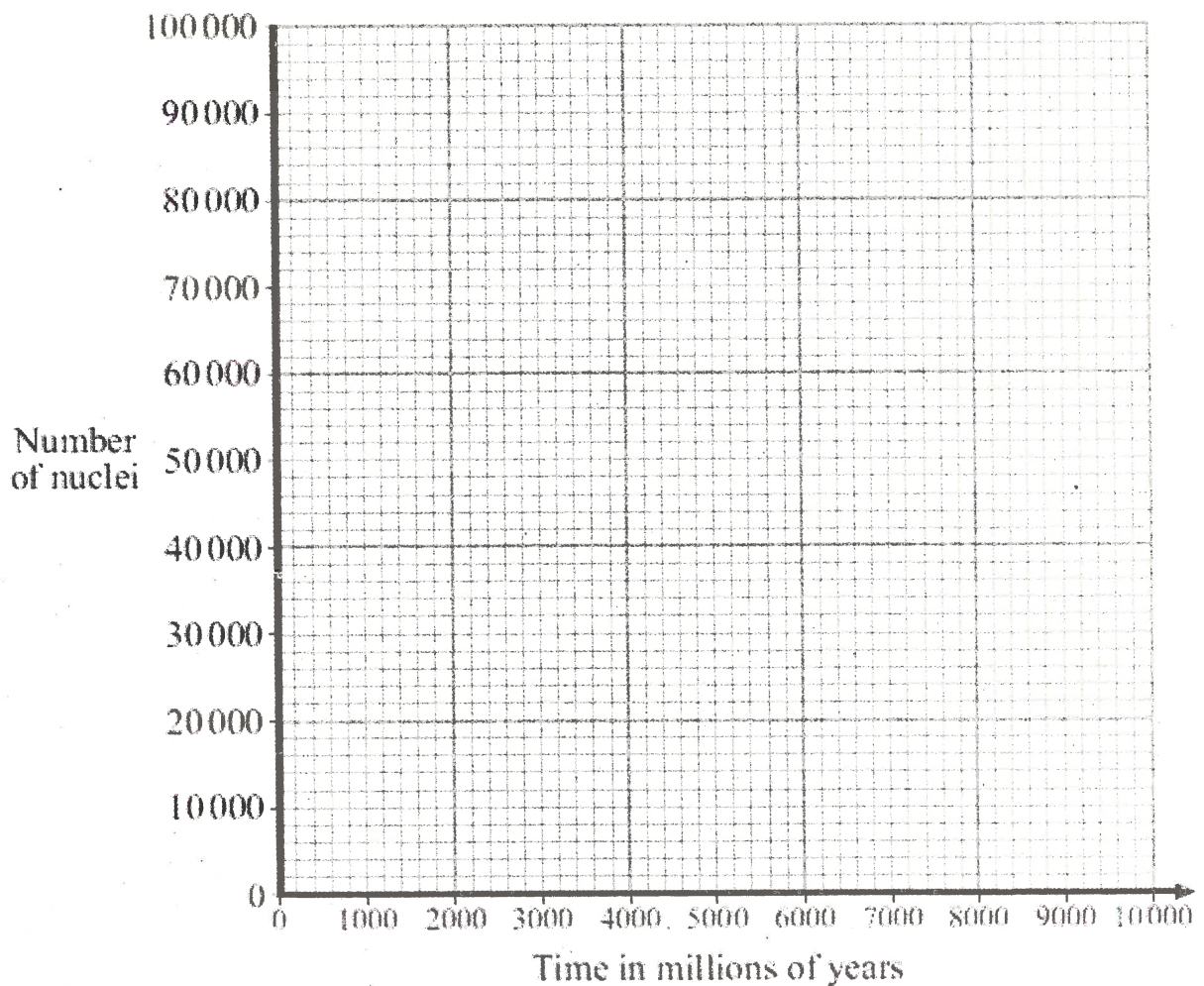
Half-life

1. What are possible units for half life?
2. A radioactive isotope has a half life of 14 days. It has an initial count rate of 1080Bq. What will the count rate be after 4 weeks?
3. A radioactive isotope has a half life of 15 minutes. It has an initial count rate of 36000 Bq. What will the count rate be after 1.5 hours?
4. A radioactive isotope has a half life of 5000 years. What fraction of the radioactive material will remain after 20000 years?
5. A radioactive isotope has a count rate of 4000Bq and a half life of 12 hours. How long will it take the count rate to drop to 500Bq?
6. Use the graph to find the half-life of Plutonium-238.



Further Questions:

7. A radioactive isotope has a count rate of 6400 Bq and a half life of 4 days. What fraction of the isotope will have decayed after 20 days? What will the count rate now be?
8. Before an isotope is placed near the detector a counter gives a reading of 14 Bq. When the isotope is placed near the reading increases to 2234 Bq. When tested 30 minutes later the count rate has dropped to 569 Bq. What is the half life of the substance?
9. Uranium-238 has a half-life of 4500 million years. Complete the graph to show the number of nuclei in a sample of U-238 will change over time. Initially there are 100,000 nuclei in the sample.



Name: _____

Date: _____

Half - Life Practice

1. What is meant by half-life?

2. If you have 100 grams of a radioactive isotope with a half-life of 10 years:
 - a. How much of the isotope will you have left after 10 years?

 - b. How much of the isotope will you have left after 20 years?

 - c. How many half-lives will occur in 40 years?

3. The half-life of plutonium-239 is 24,300 years. If a nuclear bomb released 8 kg of this isotope, how many years would pass before the amount is reduced to 1 kg?

4. The half-life of radon-222 is 3.8 days. How much of a 100 gram sample is left after 15.2 days?

5. Carbon-14 has a half-life of 5,730 years. If a sample contained 70 mg originally, how much is left after 17,190 years?

6. The half-life of cobalt-60 is 5.26 years. If 50 grams are left after 15.78 years, how many grams were in the original sample?

7. The half-life of I-137 is 8.07 days. If 25 grams are left after 40.35 days, how many grams were in the original sample?

8. If 100 grams of Au-198 decays to 6.25 grams in 10.8 days, what is the half-life of Au-198?

Radioactive decay: Worksheet 4.2.3

Key terms

1. Give a definition of each of the following key words.

Radioisotope:

Alpha particle:

Beta particle:

Gamma ray:

Activity of radioactive sources

The activity of a radioactive source is the number of nuclear decays in one second. Activity is measured in becquerels (Bq) or counts per second.

1 Bq = 1 count per second

2. The activity of a radioactive source is 150 Bq. Assuming that all decays are counted, how many counts would be recorded in:
 - (a) 1 s?
 - (b) 20 s?
 - (c) 60 s?
3. An activity of 200 Bq was recorded for a radioactive source. How many counts would be recorded in:
 - (a) 2 min? (1 min = 60 s)
 - (b) 1 hour? (1 hour = 60 min)
 - (c) 24 hours?

Penetrating power

4. Link the type of radiation to the substance it can be stopped by.

Type of radiation	Stopped by
Alpha	Thick lead or concrete
Beta	Sheet of aluminium
Gamma	Sheet of paper

Types of decay

5. When a nucleus undergoes radioactive decay, the mass number is unchanged but the atomic number has increased by one. What type of decay is this?

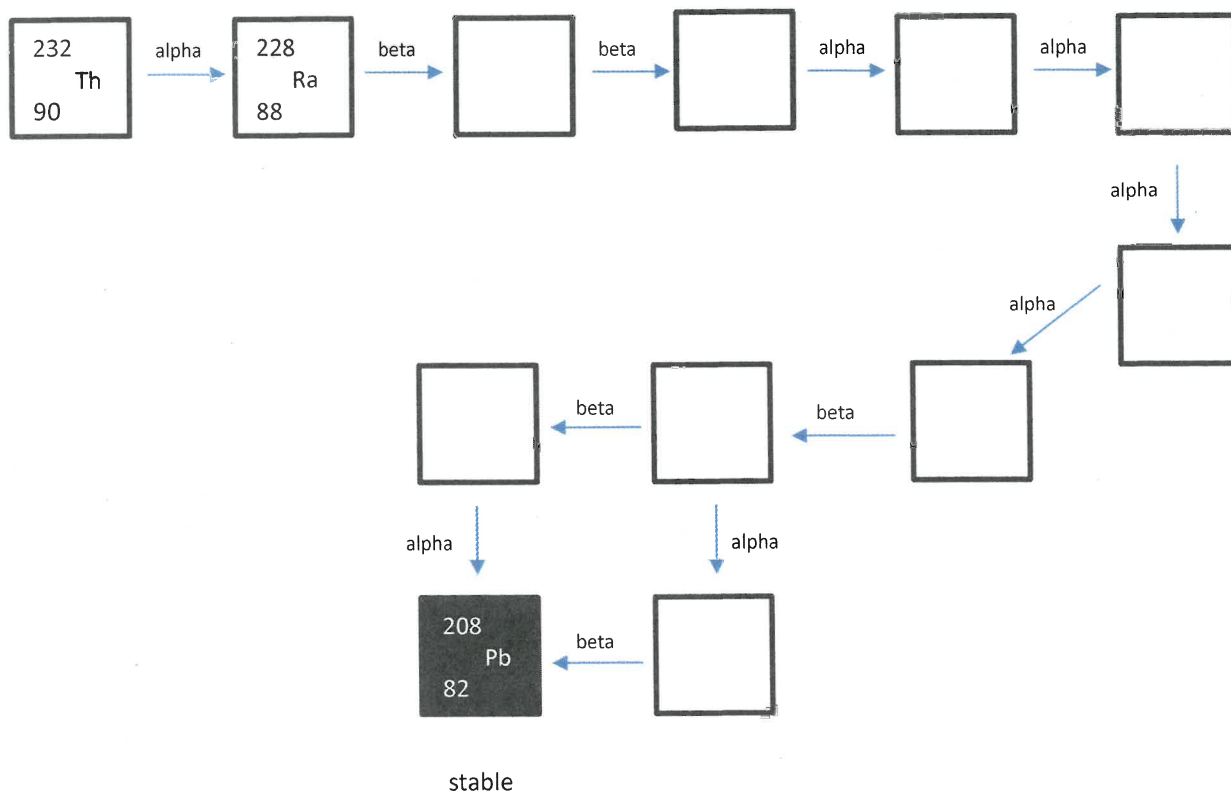
Nuclear equations: Worksheet 4.4.3

Introduction

A decay chain is a sequence of nuclear equations that shows how unstable nuclei decay until a stable atom, usually lead, is reached.

Task

Write the resultant (daughter) nuclei in this decay chain, given the decay process outlined on the arrows. The first one has been done for you.



Radioactive half-life: Worksheet 4.5.3**Introduction**

The activity of a radioisotope can be measured using a meter. The meter records the activity of the radioisotopes in counts per second. The data collected can be used to determine the half-life of the radioisotope. This is useful because some applications require a short half-life (such as medical tracers), whereas other applications require use of a radioisotope with a much longer half-life (such as smoke alarms).

Half-life of a radioisotope

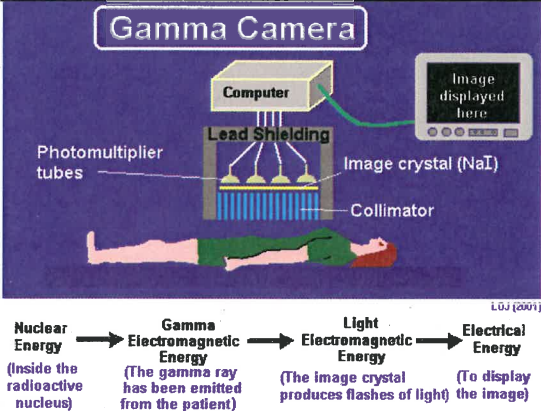
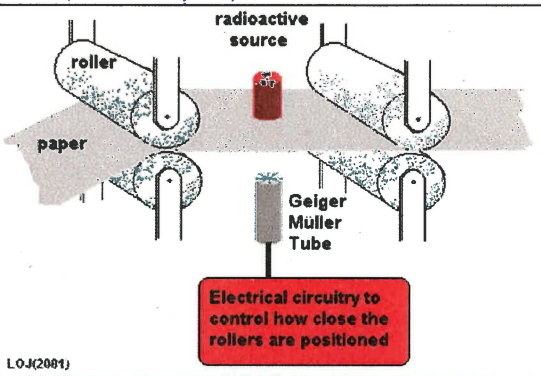
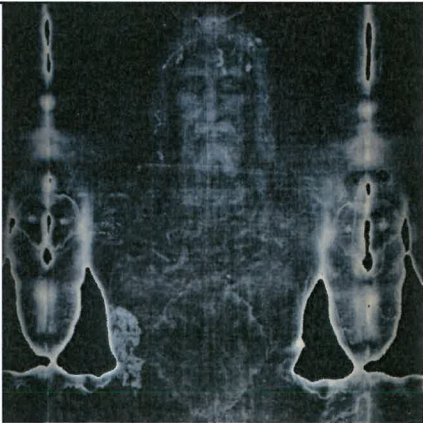
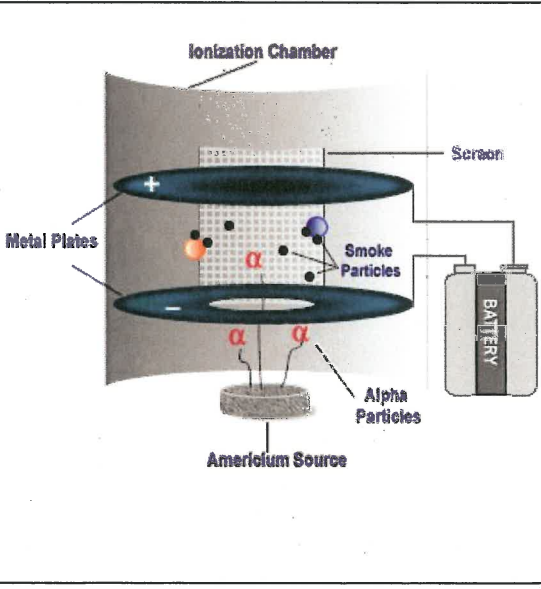
The activity of a radioisotope was measured every 10 minutes for an hour. The following results were collected.

Time (minutes)	Count rate (counts per second)
0	700
10	520
20	400
30	310
40	245
50	175
60	150

1. Plot a graph of the data provided. Label each axis on the graph with a quantity and a unit. Add a smooth curve of best fit through the points.
2. By taking several readings from the graph, determine the half-life of the radioisotope.

Uses of Radioactive Sources

Match the use with its description on the right.

Radioactive Tracer		Used to find the age of ancient materials. Living wood has a tiny proportion of radioactive carbon C-14.
Thickness Control in Manufacturing		A small amount of radioisotope is made to replace a non-radioactive isotope of an element in a compound. The path of that radioisotope through the patient is then monitored by detection of emitted nuclear radiation.
Carbon dating		Some smoke detectors contain a small amount of Americium-241, an alpha emitter with a half life of 460 years. Smoke entering the detector blocks some of the alpha particles, triggering the alarm.
Smoke Detectors		Automatic control over the thickness of paper in paper mills can be obtained by passing beta radiation through the paper and monitoring the count rate. An isotope with a long half-life is used so that the count-rate hardly changes with time. A constant rate is maintained. If the rate is too low the rollers automatically move closer to each other (making the paper thinner) and vice versa.